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Anti-biofouling marine diterpenoids from Okinawan soft corals

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ABSTRACT

Soft corals produce a diverse range of natural products with pharmaceutical potential, such as antiproliferative and anti-inflammatory effects. The Alcyoniidae family, particularly the genera *Sarcophyton* and *Sinularia*, is rich in bioactive terpenoids. However, despite extensive research, their anti-biofouling properties against the mussel *Mytilus galloprovincialis* remain underexplored. This study investigates these compounds as potential eco-friendly antifouling agents. A new cembrane-type diterpenoid, 11,12-epoxycembrene A (**1**), and 15 known compounds were isolated from three soft corals distributed in Okinawa, Japan. The chemical structures of these secondary metabolites were elucidated based on spectroscopic analysis. Moreover, an anti-biofouling assay of potential anti-biofouling agents against *M. galloprovincialis* was performed and their toxicities were assessed by means of the brine shrimp mortality test. In conclusion, this study identifies new and known bioactive compounds from soft corals, introduces an improved anti-biofouling assay, and highlights the potential of dimethylamine-containing diterpenes as environmentally friendly antifouling agents.

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Soft coral; dimethylamine; anti-biofouling assay; brine shrimp lethality test

Introduction

More than 5,000 marine natural products have been isolated from soft corals over the past 50 years (Lyu et al. 2021). Soft corals use these natural products as chemical defense agents against predators and micro-organisms and as allelopathic agents that enhance their ability to survive in competitive environments (Coll et al. 1982; La Barre et al. 1986; Changyun et al. 2008; Eskander et al. 2018). Natural products derived from soft corals are known to have pharmaceutical value and have been extensively studied. Recently, several studies have been conducted on their antiproliferative effects on oral cancer cell lines, anti-inflammatory effects on RAW264.7 macrophages, and inhibition of the STING signaling pathway in living mammalian cells (Cheng et al. 2023; Hsiao et al. 2023; Peng et al. 2023). Thus, several unknown biologically active compounds can be obtained from natural products derived from soft corals.

Chemical studies on the coral family Alcyoniidae have mostly been conducted in the South China Sea, Taiwan, Japan, Malaysia, India, and Australia (Lyu et al. 2021). Soft corals belonging to the genera

Sarcophyton and *Sinularia* from the family Alcyoniidae possess an abundance of natural products, with 487 and 797 reported compounds, respectively (Lyu et al. 2021). One of the first natural products discovered in these soft corals was cembrane diterpenoids, including sarcophine isolated from *Sarcophyton glaucum* (Bernstein et al. 1974) and sinulariolide from *Sinularia flexibilis* (Tursch et al. 1975). Since then, the chemical investigation of these soft corals has been substantially expanded to other terpenoids, such as sesquiterpenes and tetraterpenes from *Sarcophyton* species (Elkhawas et al. 2020) and sesquiterpenes from *Sinularia* species (Chen et al. 2012; Yan et al. 2021), as shown in Figure 1.

Previously, we showed that the soft coral genera *Sarcophyton* and *Sinularia* contain a diverse range of bioactive terpenoids (Kamada et al. 2016; Phan et al. 2016; Kamada et al. 2018a; Kamada et al. 2018b; Kamada et al. 2018c; Phan et al. 2019; Tani et al. 2019; Ishii et al. 2022). Our continuing search for bioactive natural products from *Sarcophyton* and *Sinularia* species collected in Okinawa, Japan, has led to the discovery of a new analog of cembrane

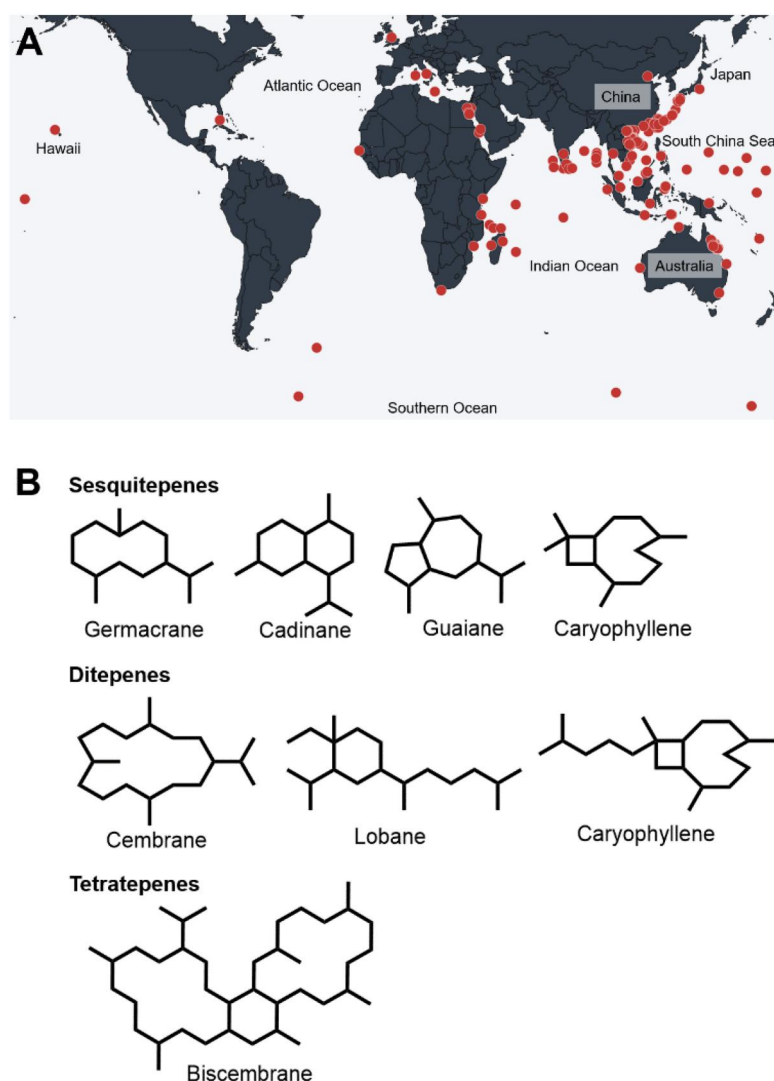


Figure 1. (A) Map of the chemical investigation of the coral family Alcyoniidae (red dots represent sampling locations). (B) Carbon skeletons of terpenes from the soft coral genera *Sarcophyton* and *Sinularia*.

diterpenoid, 11,12-epoxycembrene A (1). In addition, 15 known compounds, namely epoxyisoneocembrene A (2) (Bowden et al. 1983), sarcophytonin A (3) (Grote et al. 2008), (2S,11S,12S)-isosarcophytoxide (4) (Bowden et al. 1987), (2S,11R,12R)-isosarcophytoxide (5) (Bowden et al. 1987), 15-hydroxycembra-1,3,7,11-tetraene (6) (Bowden et al. 1987), cembranolide A (7) (Kobayashi et al. 1987), cembranolide B (8) (Kobayashi et al. 1987), (1S,2S,3E,7E,11E)-3,7,11,15-cembratetraen-17,2-olide (9) (Kobayashi et al. 1987), sinulariol A (10) (Kobayashi et al. 1987), mayol (11) (Uchio et al. 1981), denticulatolide (12) (Kobayashi et al. 1987), lobophytol (13) (Iguchi et al. 1991), lobophytol acetate (14) (Iguchi et al. 1991), mayolide A (15) (Kobayashi 1988), and sinulamine I (16) (Iguchi et al. 1992) were identified (Figure 2). In this study, we report the isolation and structural elucidation of these isolated compounds.

Biofouling is a global issue, creating a demand for environmentally friendly antifouling agents. However, reports on the anti-biofouling effects of such agents against the mussel *Mytilus galloprovincialis* are scarce. Therefore, we also conducted an anti-biofouling assay against *M. galloprovincialis* (Ishigami et al. 2024) and evaluated the toxicity using the brine shrimp lethality test (Ishii et al. 2017). In addition, a previously reported method (Ishigami et al. 2024) encountered problems with the byssus threads sticking to the outside of the tested zone. However, expanding the test zone requires more samples. This method is not suitable for biological assays with natural products. In this study, the antifouling assay method was improved to solve this problem and search for safer and more environmentally friendly marine-derived coatings. Some extracts of soft corals have shown antifouling effects, potentially alternating current bioassay methods

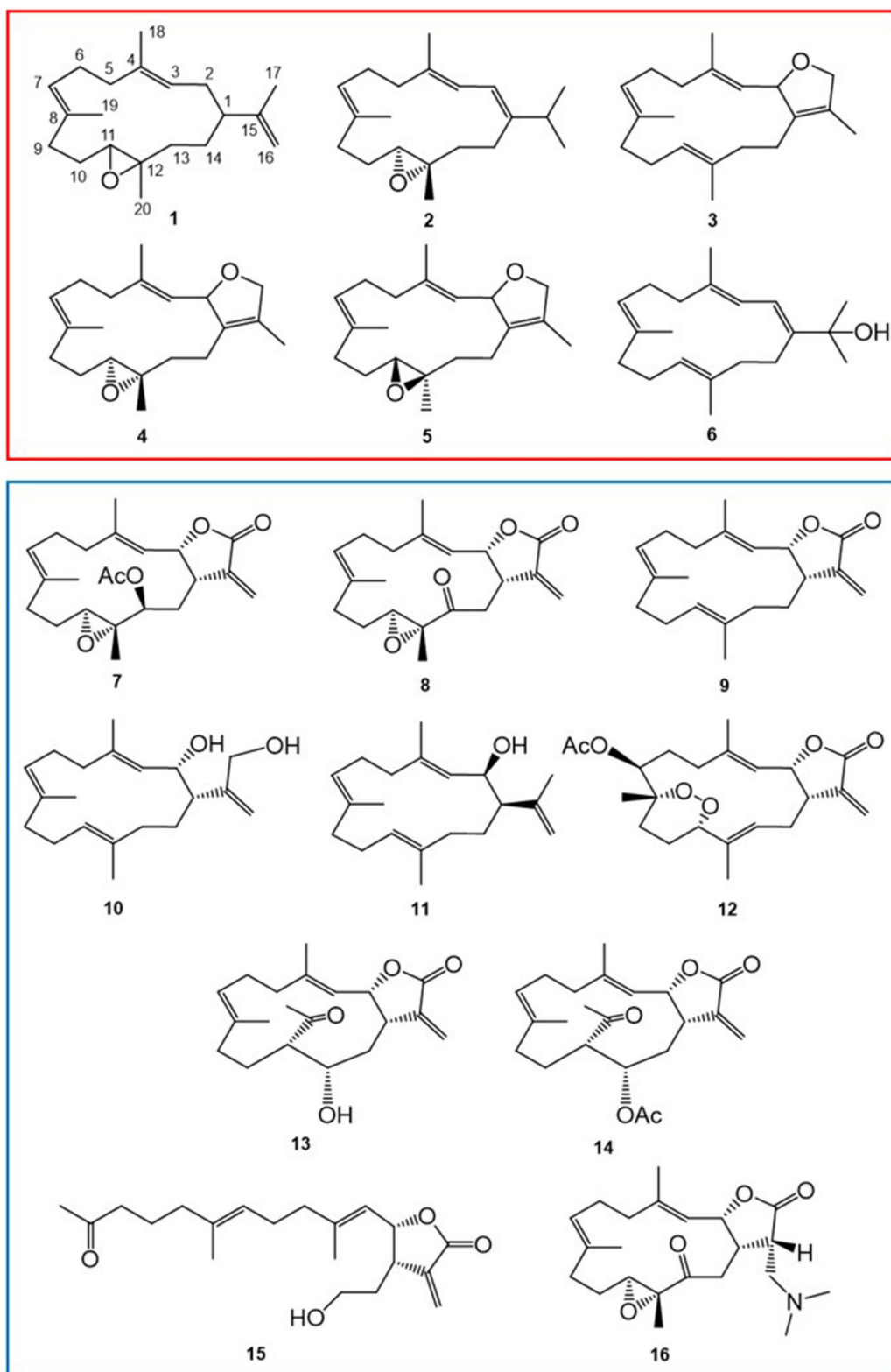


Figure 2. Structures of compounds 1–16. Compounds obtained from *Sarcophyton glaucum* are red and the ones from *Sinularia* sp. are blue.

and reducing toxic impacts in the ocean. This study confirms the anti-biofouling activity of several cem-brane-type diterpenoids. These findings highlight the

potential of marine soft corals as a natural source of bioactive compounds with significant anti-biofouling properties. Furthermore, the effectiveness of such

compounds underscores the importance of exploring marine resources for developing environmentally friendly alternatives to address the issues associated with biofouling.

Materials and methods

General experimental procedures

The optical rotations of the compounds were measured using a Jasco P-1010 digital polarimeter (Jasco, Tokyo, Japan). ^1H - (400 MHz) and ^{13}C -NMR (100 MHz) spectra were recorded on 400 MHz FT-NMR (JEOL, Tokyo, Japan) using chloroform-*d* (CDCl_3) as the solvent and tetramethyl silane as the internal standard. Infrared (IR) spectra were recorded using an FT/IR-6100 spectrometer (Jasco). High-resolution ESI mass spectra (HR-ESI-MS) were obtained using a SYNAPT HDMS (Waters, Milford, Massachusetts, USA). Preparative thin-layer chromatography (PTLC) was performed using silica gel glass plates (Merck, Darmstadt, Germany; Kieselgel 60 F₂₅₄), and column chromatography was performed using silica gel (Merck, Kieselgel 60, 70–230 mesh). A new webbron color (Tokushu Tokai Paper, Tokyo, Japan) was used for the anti-biofouling assay. Dried brine shrimp eggs (Tetra, Melle, Germany) and Hatcher 24 (Nippon Zoologic, Tokyo, Japan) were used for the brine shrimp lethality test.

Biological materials

Specimens of the soft corals *S. glaucum* and *Sinularia* sp. were collected from the coasts of Minato town, Naha, Okinawa, on 2 March 2022, whereas those of the soft coral *Sinularia* sp. were collected from the Odo Beach, Itoman town, Okinawa, on 28 February 2021. The extract was stored at -40°C until further analysis.

Extraction and isolation

Air-dried *S. glaucum* from Minato town, Naha, Okinawa, was extracted with methanol (MeOH) at 22°C for two days. This process was repeated three times. The crude MeOH extract was suspended in distilled water (1.0 L) and partitioned with ethyl acetate (EtOAc, 1.5 L). After removing the organic solvent, the crude EtOAc extract (1.5 g) was chromatographed on a silica gel column using *n*-hexane and EtOAc as the eluents with increasing polarity (*n*-hexane/EtOAc = 9:1, 4:1, 7:3, 1:1, and EtOAc) to yield five fractions. Fraction 2 (200 mg/420 mg) was

subjected to PTLC using toluene and yielded five fractions. Fraction 2-3 was subjected to PTLC with *n*-hexane/EtOAc (97:3) to yield five fractions, including compound **3** (9.7 mg). Fraction 2-3-2 was subjected to PTLC with benzene to obtain compounds **1** (2.0 mg) and **2** (6.5 mg). Fraction 2-4 was subjected to PTLC using *n*-hexane/EtOAc (95:5) to obtain compound **6** (21.5 mg). Fraction 3 (300 mg/878 mg) was subjected to PTLC using CHCl_3 to yield five fractions. Fraction 3-2 was subjected to PTLC using *n*-hexane/EtOAc (95:5) to obtain compound **4** (3.6 mg). Fraction 3-3 was subjected to PTLC using *n*-hexane/EtOAc (8:2) to obtain compound **5** (5.0 mg).

Air-dried *Sinularia* sp. from the Odo coast, Itoman, Okinawa, was extracted and partitioned as previously described. The EtOAc crude extract (1.5 g/3.6 g) was chromatographed on a silica gel column using *n*-hexane and EtOAc as eluents with increasing polarity (*n*-hexane/EtOAc = 9:1, 4:1, 7:3, 1:1, and EtOAc) to yield five fractions. Fraction 2-4 was subjected to PTLC using *n*-hexane/EtOAc (9:1) to yield five fractions. Fraction 2-4-4 was subjected to PTLC using CHCl_3 to yield five fractions. Fraction 2-4-4-2 was subjected to PTLC using *n*-hexane/EtOAc (9:1) to obtain compounds **9** (9.5 mg) and **11** (3.6 mg). Fraction 3 (100 mg/400 mg) was subjected to PTLC using CHCl_3 to yield five fractions. Fraction 3-2 was subjected to PTLC with *n*-hexane/EtOAc (8:2) to yield four fractions. Fraction 3-2-2 was subjected to PTLC using toluene/EtOAc (95:5) to obtain compound **7** (4.7 mg). Fraction 3-2-3 was subjected to PTLC using benzene to yield five fractions. Fraction 3-2-3-5 was subjected to PTLC using toluene/EtOAc (95:5) to obtain compounds **8** (3.1 mg) and **12** (10.0 mg). Fraction 4 (100 mg/512 mg) was subjected to PTLC using *n*-hexane/EtOAc (5:2) to obtain compounds **13** (10.0 mg) and **14** (49.4 mg). Fraction 4-5 was subjected to PTLC using *n*-hexane/EtOAc (3:1) to obtain compound **10** (1.2 mg). Fraction 5 (100 mg/117 mg) was subjected to PTLC using *n*-hexane/EtOAc (3:2) to yield five fractions. Fraction 5-4 was subjected to PTLC using $\text{CHCl}_3/\text{MeOH}$ (9:1) to yield five fractions. Fraction 5-4-2 was subjected to PTLC with toluene/EtOAc (2:1) to obtain compound **15** (10.0 mg).

Air-dried *Sinularia* sp. from Minato town, Naha, Okinawa, was treated as described above. The EtOAc crude extract (3.19 g/3.37 g) was chromatographed on a silica gel column using *n*-hexane and EtOAc as eluents with increasing polarity (*n*-hexane/EtOAc = 9:1, 4:1, 7:3, 1:1, EtOAc, and MeOH) to yield six fractions. Fraction 2 (100 mg/321 mg) was subjected to PTLC

using toluene, which yielded five fractions. Fraction 2-3 was subjected to PTLC using *n*-hexane/EtOAc (17:2) to obtain compounds **9** (5.3 mg) and **11** (54.3 mg). Fraction 2-5 was subjected to PTLC using *n*-hexane/EtOAc (17:2) to obtain compound **7** (3.9 mg). Fraction 3 (100 mg/789 mg) was subjected to PTLC with *n*-hexane/EtOAc (8:2) to obtain compounds **8** (18.0 mg) and **12** (27.1 mg). Fraction 4 (100 mg/958 mg) was subjected to PTLC with *n*-hexane/EtOAc (8:2) to obtain compounds **13** (4.0 mg) and **14** (65.8 mg). Fraction 5 (100 mg/523 mg) was subjected to PTLC using *n*-hexane/EtOAc (1:1) to obtain compound **15** (21.6 mg). Fraction 6 (788.9 mg) was subjected to PTLC using CHCl₃/MeOH (95:5) to yield four fractions. Fraction 6-1 was subjected to PTLC twice with *n*-hexane/EtOAc (1:1) to obtain compound **16** (6.2 mg).

Characterization of 11,12-epoxycembrene A (**1**)

Colourless oil; $[\alpha]_D^{21}$ of -27.0 (c 0.1, CHCl₃); ¹H- and ¹³C-NMR spectral data (Table 1). HR-ESI-MS: m/z 289.2523 $[M + H]^+$ (calculated for C₂₀H₃₃O, 289.2526).

Anti-biofouling assay

The test methods used were those reported (Ishigami et al. 2024). *M. galloprovincialis* used in the study were collected from the shore of Kesenuma, Miyagi, Japan, and biofouling prevention assays were performed at 0.16 μmol/cm² of each compound. Acrylic

plate (20 cm × 40 cm) and new webron color [two sheets: 4 cm × 20 cm (A), one sheet: 6 cm × 20 cm (B), two sheets: 2 cm × 20 cm (C), four sheets: 1 cm × 20 cm (D)] were prepared for the anti-biofouling assay. A and B were combined with copper sulfate pentahydrate (CuSO₄·5H₂O) and soaked in an aqueous solution saturated with CuSO₄·5H₂O for 10 min and dried for 30 min. They were arranged in the order of A, C, B, C, and A and glued to an acrylic plate. Next, one piece of D with a sample applied and one piece of D' with no application (serving as blank zones) were glued together. Five mussels were glued with a rubber piece to the border between D and D', resulting in ten test zones. The number of byssus threads in the sample (D) and blank zones (D') were calculated using the following formula: Inhibition rate (%) = $\{D'/(D'+D)\} \times 100$ (Figure 3).

Brine shrimp toxicity assay

The test methods used were those reported (Ishii et al. 2017). To incubate brine shrimp, 450 mL of water was placed in a container. To this, 13 g of salt was added, stirred well, and brine shrimp eggs (0.5 g) were added. The water temperature was adjusted to 27 °C in a hot water bath and aeration was performed. After 24 h, 20 eggs were collected from the hatched brine shrimp using a dropper and removed to 24 well plates (20 eggs per well). The brine shrimp removed to 24 well plates were transferred to a measuring cylinder, and seawater was added up to the 4 mL mark. Next, 0.05 mg of each sample was weighed into a vial, and 100 μL of dimethyl sulfoxide (DMSO) was added. To this, 900 μL of seawater was added, followed by 4 mL of seawater containing 20 brine shrimps. The vials were then placed in a hot water bath at 27 °C. After 24 h, the number of live brine shrimp (A: Alive) and lethal (D: Dead) were counted. The lethality was calculated using the following formula: lethality (%) = $\{D/(A + D)\} \times 100$.

Results and discussion

Compound **1** was obtained as a colorless oil with a $[\alpha]_D^{21}$ of -27.0 (c 0.1, CHCl₃). The molecular formula, C₂₀H₃₂O, was determined by HR-ESI-MS with a positive ion at m/z 289.2523 $[M + H]^+$ (calculated for 289.2526). The ¹H- and ¹³C-NMR spectra (Table 1) revealed twenty carbon signals that were assigned using heteronuclear single quantum coherence (HSQC). These signals corresponded to four methyl groups, eight methylenes, four methines, and four quaternary carbon

Table 1. ¹H- and ¹³C-NMR data (400 and 100 MHz) for compound **1** in CDCl₃.

Compound 1		
Position	δ _C	δ _H (J, Hz)
1	46.9	1.97 (1H, m)
2	30.9	2.08 (2H, m)
3	123.4	5.08 (1H, m)
4	135.8	
5	39.2	2.17 (2H, m)
6	24.9	2.20 (2H, m)
7	126.4	5.05 (1H, m)
8	133.2	
9	36.9	2.24 (1H, m)
		2.12 (1H, m)
10	24.4	2.01 (1H, m)
		1.34 (1H, m)
11	62.7	2.57 (1H, dd, $J = 9.8, 3.0$)
12	62.4	
13	37.7	1.84 (1H, m)
		0.83 (1H, m)
14	26.9	1.46 (1H, m)
		1.32 (1H, m)
15	148.3	
16	110.5	4.76 (1H, m)
		4.68 (1H, m)
17	20.3	1.69 (3H, s)
18	15.6	1.54 (3H, s)
19	15.0	1.61 (3H, s)
20	16.6	1.02 (3H, s)

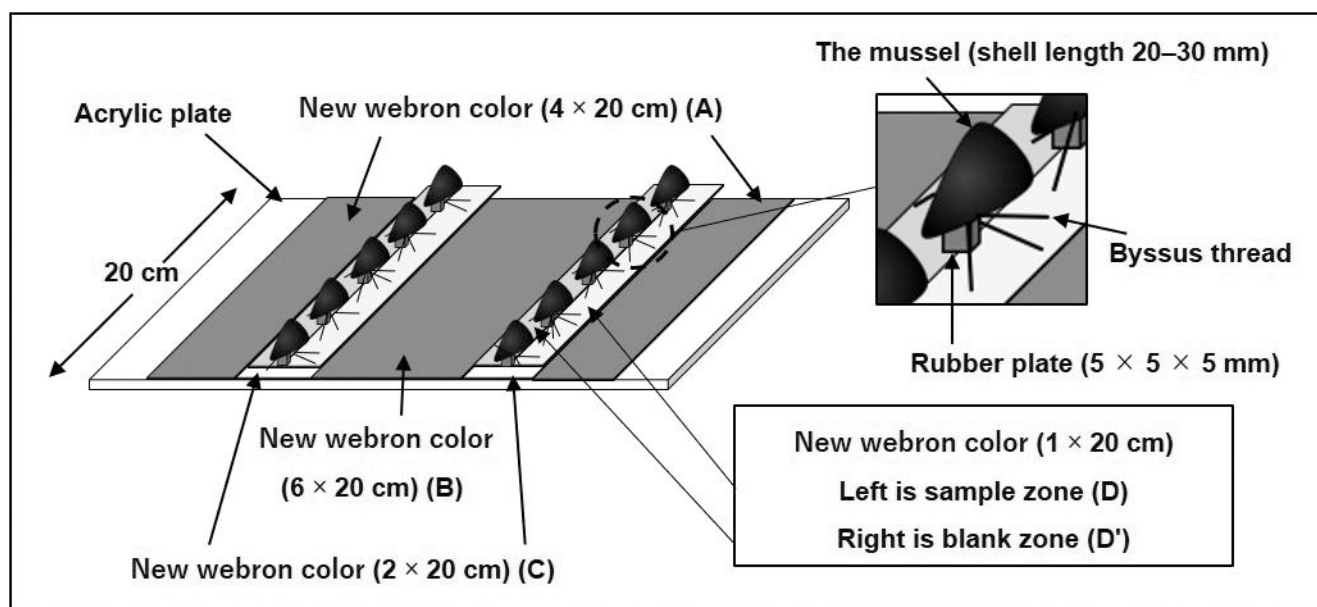


Figure 3. Modified anti-biofouling assay for *Mytilus galloprovincialis* mussels.

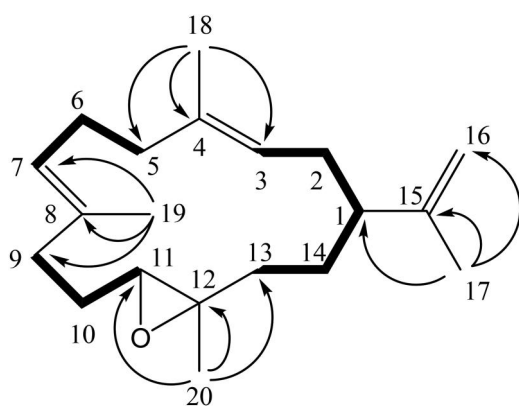


Figure 4. ^1H - ^1H COSY (bolds) and key HMBC (arrows) correlations of **1**.

atoms. Six olefinic carbons (δ_{C} 148.3, 135.8, 133.2, 126.4, 123.4, and 110.5) and two oxirane carbons (δ_{C} 62.7 and 62.4) were observed. Furthermore, three pairs of double bonds accounted for three of the five degrees of unsaturation deduced from the molecular formula, indicating the presence of two monocyclic systems in **1**. The ^1H -NMR spectrum revealed the presence of four singlet methyls (δ_{H} 1.69, 1.61, 1.54, and 1.02) and one olefinic methylene (δ_{H} 4.76 and 4.68). A detailed analysis of the ^1H - ^1H correlation spectroscopy (COSY) spectrum of **1** allowed the construction of three spin systems, namely C3-C2-C1-C14-C13, C5-C7, and C9-C11. The heteronuclear multiple bond correlation (HMBC) spectrum showed correlations between H_3 -17/C1, C15, and C16; H_3 -18/C3, C4, and C5; H_3 -19/C7, C8, and C9; and H_3 -20/C11, C12, and C13 (Figure 4). Based on these findings, the planar structure of **1**

was elucidated as 11,12-epoxycembra-3,7,15-trien. Unfortunately, the stereochemistry of **1** could not be discussed because appropriate nuclear overhauser effect spectroscopy (NOESY) data was not obtained due to the limited amount of **1**. As of 25 July 2024, a SciFinder search for the chemical structure of **1** yielded only one relevant reference (Kuznetsova et al. 1989). However, the chemical structure in that reference differed from that of the identified compound, namely 11,12-epoxycembra-3,7,15-trien-13-ol or 11,12-epoxycembra-3,7,15-trien-13-one (Kuznetsova et al. 1989). Therefore, this study is the first to investigate compound **1**. The other compounds were identified as epoxyisoneocembrene A (**2**) (Bowden et al. 1983), sarcophytonin A (**3**) (Grote et al. 2008), (2S,11S,12S)-isosarcophytoxide (**4**) (Bowden et al. 1987), (2S,11R,12R)-isosarcophytoxide (**5**) (Bowden et al. 1987), and 15-hydroxycembra-1,3,7,11-tetraene (**6**) (Bowden et al. 1987).

Compounds **7**-**15** were obtained from the soft coral *Sinularia* sp. from the Odo coast, Itoman, Okinawa. A comparison of the spectroscopic data with published papers revealed that these compounds were cembranolide A (**7**) (Kobayashi et al. 1987), cembranolide B (**8**) (Kobayashi et al. 1987), (1S,2S,3E,7E,11E)-3,7,11,15-cembratetraen-17,2-olide (**9**) (Kobayashi et al. 1987), sinulariol A (**10**) (Kobayashi et al. 1987), mayol (**11**) (Uchio et al. 1981), denticulatolide (**12**) (Kobayashi et al. 1987), lobophytol (**13**) (Iguchi et al. 1991), lobophytol acetate (**14**) (Iguchi et al. 1991), and mayolide A (**15**) (Kobayashi 1988).

Table 2. Anti-biofouling activity of tested compounds.

Compound	Inhibition rate (%) (0.16 $\mu\text{mol}/\text{cm}^2$)
$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$	90
1	58
2	<50
3	<50
4	64
5	58
6	<50
7	83
8	64
9	88
10	N/A
11	83
12	51
13	67
14	79
15	55
16	81

Furthermore, compounds **7–9**, **11–15**, and **16** were obtained from the soft coral *Sinularia* sp. from Minato town, Naha, Okinawa. Compound **16** was identified as sinulamine I (Iguchi et al. 1992). Dimethylamine-containing diterpenes from soft corals are rare, and previous research has only confirmed their HIV-inhibitory activity (Rashid et al. 2000).

An anti-biofouling assay was performed on the mussel *M. galloprovincialis*. The assay was a modified version of a previously reported method (Ishigami et al. 2024). Applying the compound over a wide area to limit the site of byssus thread formation resulted in the use of a large amount of the compound, which was inappropriate for a natural product that could only be obtained in limited quantities. In this method devised this time, by using $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ over a wide area to limit the byssus thread formation site, the byssus thread attachment site of the mussel was successfully limited to a narrow area, saving approximately 14 mg per compound. The specific method was described in the previous section. The anti-biofouling activities of the tested compounds are listed in Table 2. Compounds **7**, **9**, **11**, and **16** showed strong activities above 80%. In particular, compound **9** showed activity as strong as that of the positive control, the copper compound ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$). Therefore, these compounds have potential as antifouling agents. The brine shrimp toxicity assay was conducted as previously described (Ishii et al. 2017). The brine shrimp lethality of the tested compounds is shown in Table 3. The lethality rates of all compounds except **11** were below 25%, suggesting that these compounds tend to be less toxic to small crustaceans. Further investigation with cyanobacterial and microalgal models as well as fish is required to elucidate the ecotoxicity of them.

Table 3. Brine shrimp lethality of tested compounds.

Compound	Lethality (%) (10 $\mu\text{g}/\text{mL}$)
Laurinterol	100
1	22
2	5
3	17
4	5
5	20
6	17
7	13
8	13
9	25
10	N/A
11	65
12	8
13	5
14	0
15	5
16	5

The true producers of marine natural products have been debated. A notable example is polytheonamides, which were originally thought to be isolated from the marine sponge *Theonella swinhoei* (Hamada et al. 2005). However, it was discovered that these compounds are actually produced by the sponge-associated bacteria *Entotheonella* (Freeman et al. 2012). Thus, sponge-derived natural products, such as polytheonamides, originate from their microbiomes. Similar to sponges, the origin of soft coral-derived natural products has remained intractable. Recent studies on soft corals have indicated that the origin of diterpenes is from the host organism by characterizing soft coral-encoded terpene cyclases, which are involved in diterpene biosynthesis (Burkhardt et al. 2022; Scesa et al. 2022). This has had a significant impact on biomedical applications by exploiting the biotechnological potential of soft corals, such as the gram-scale production of drug-like scaffolds and eunicellane diterpenoids by coral terpene cyclases combined with a semisynthetic strategy (Scesa et al. 2023). The biotechnological potential of soft corals was first initiated by chemical investigation, followed by genomics. Our chemical investigation expanded the chemical diversity of diterpenoids. In addition, compound **16** features a unique dimethylamine group at C-16. To date, only four examples of cembrane-type diterpenoids with dimethylamine group at C-16 have been reported from soft corals (Iguchi et al. 1992; Rashid et al. 2000; Chao et al. 2008). This highlights the unknown soft coral-encoded enzymes that catalyze the dimethylamine moiety. Anti-biofouling assay results showed that the repellencies of cembranolide B (**8**) and sinulamine I (**16**) were 64% and 81%, respectively. Compound **16** was 17% more active than **8**, which did not contain a dimethylamine group. This

suggested that the dimethylamine group was an important structure that improved the anti-biofouling activity. In future, the optimal structure for anti-biofouling activity can be elucidated by synthesizing compounds containing dimethylamine groups and clarifying the activity relationships.

Conclusions

We obtained a new compound, 11,12-epoxycembrene A (**1**), and 15 known compounds, including sinulamine I (**16**), a rare dimethylamine-containing diterpene, from three populations of soft corals collected in Okinawa, Japan. Subsequently, we modified and examined an anti-biofouling bioassay against blue mussels. Compounds **7**, **9**, **11**, and **16** exhibited strong activities above 80%. In addition, compound **16** had a low ecotoxicity (5%) in a brine shrimp lethality test. When compared to compound **8**, which has a similar chemical structure but lacks a dimethylamine group, compound **16** exhibited stronger anti-biofouling activity (17% higher inhibition rate to blue mussel) and lower ecotoxicity (8% less lethality to brine shrimp). This raised expectations regarding the role of the dimethylamine group in the bioactivity. The chemical synthesis of dimethylamine-containing diterpenes, evaluation of the biological activity of their synthetic derivatives, and elucidation of the biosynthetic pathways and enzymes that produce dimethylamine groups under natural conditions may contribute to the development of new antifouling agents in the future.

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Authors contributions

Writing the original draft: SI, RN, and CSP. Methodology: SI, AS, FY, HT, KN, and IO. Data curation and formal analysis: SI and TK. Writing, review, and editing: CSP and TI. Project administration and supervision: TK and TI.

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Data availability statement

All relevant data is within the manuscript.

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